UNIVERSITY OF CAMBRIDGE

LLM: FinBERT

Focus of the day: FinBERT- A Large Language Model for Extracting Information from Financial Text

As the lecture is based on the original paper, students are advised to read *FinBERT-A Large Language Model for Extracting Information from Financial Text* by Huang, Wang and Yang 2022

For the live tutorial, please access https://colab.research.google.com/- in the live tutorial, I will cover end-to-end deployment of FinBERT including installation of relevant libraries

LLM RECAP

LLM Recap

- Early work in applying NLP to finance has been based on algorithms using BoW type of structure (shortcomings include: ignore context, word order or grammar)
- LLMs are deep-learning-based NLP algorithms and are so called because of the large numbers of parameters used in calibration; they take into account semantic and syntactic relations among words. Examples include: OPEN AI GPT3, Google's BERT. They use "contextualised embeddings" i.e. word representation uses different vectors depending on context-it encodes syntactic and semantic knowledge
- LLMs challenges include:1) difficult and costly to train(large data set, computational time and energy);2)low interpretability; 3)generic vs. specialised language

FinBERT OVERVIEW

FinBERT- an introduction

- Goal was to assess the effectiveness of a LLM modelled using financial text in performing industry relevant tasks like:
  - Sentiment analysis
  - Topic Modelling
- This includes assessment on smaller sample sizes, that is especially relevant in a field like finance (e.g. availability and cost of manually labelled training data)

FinBERT Pre-training

- Purpose of pre-training using a large corpus of data is for a model to “learn the syntactic and semantic relations and linguistic patterns in a natural language”.
- Pre-training using three types of financial texts:
  - Corporate annual and quarterly filings (6,049 10-Ks and 142,622 10-Qs of Russell 3000 firms between 1994 and 2019 from SEC’s EDGAR website)
  - Financial analysts reports-47,633 reports issued for S&P500 firms period 2003-2012 from the Thomson Investext database
  - Earnings conference calls transcripts-136,578 transcripts of 7,740 public firms period 2004-2019 from SeekingAlpha website
- Total of 4.9 billion tokens (50% larger than the 3.3 billion tokens used by Google to pre-train BERT)

FinBERT-Testing

- Assessment focuses on sentiment classification problem:
  - Benchmarked vs LM dictionary and other NLP algorithms (e.g.: SVM and LSTM)
  - Same training and testing sample for all comparators: financial analysts reports with research labelled sentiments.
  - Sample includes 10k sentences: positive=3577; neutral=4586; negative=1837
  - The above approach is deemed less noisy than approaches that substitute labels with market reaction (price) [e.g. Frankel et a (2022) and Siano and Wysocki (2021)-use stock fundamentals and return as “sentiment labels” for 10-k/earnings announcements]
  - Classification happens at the sentence level because: a) sentence is regarded as the natural unit for expressing an opinion; b) practical point: BERT max input is 512 tokens!

FinBERT-Testing

- Sample is split as follows: 81% for training to fit the parameters of the ML, 9% for validation to set hyper-parameters and monitor model performance; 10% for testing
- For NLP algorithms, each sentence is assigned to the sentiment category with the highest likelihood assigned by the model
- For LM dictionary, classification rule is as follows: negative if one word from LM Fin-Neg, positive if no word from LM Fin-Neg and at least one word belonging to LM Fin-Pos; else, neutral

FinBERT-Testing

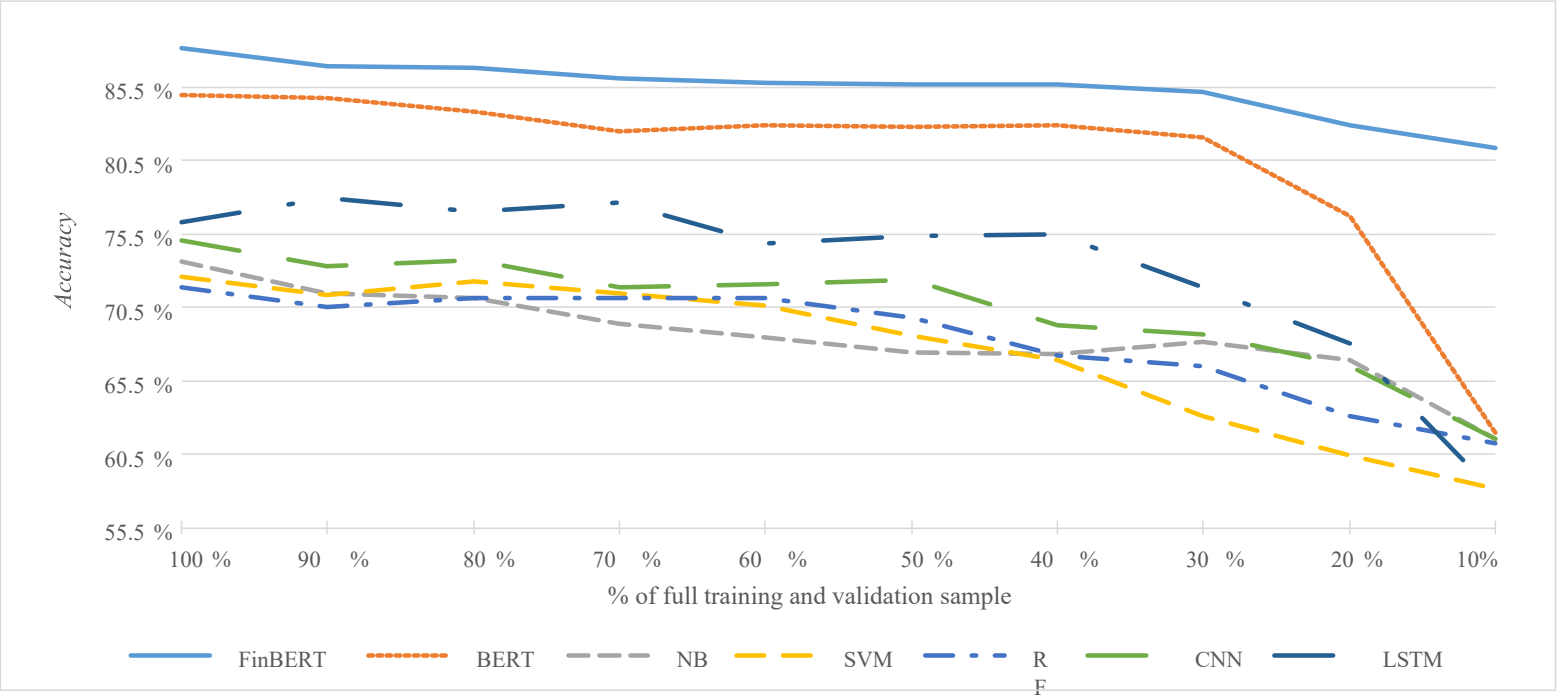
- In assessing the results, researchers use multiple measures namely:
  - *Accuracy*=the number of sentences correctly classified scaled by the total number of sentences in the testing sample
  - *Precision*=the number of sentences correctly classified into a category scaled by the total number of sentences classified into that category
  - *Mean Recall*= the number of sentences correctly classified into a category scaled by the number of sentences researchers label as belonging to that category
  - *Mean F1 score*=harmonic mean of Precision and Recall.
- Performance assessment also by sentiment type
- Assessment is also conducted with progressively smaller sample size- highly significant in practical finance applications.(Sample size is reduced incrementally down to 10% of the original set)

FinBERT-Results

		Overall (%)			Positive (358 sentences)	Neutral (458 sentences)	Negative (184 sentences)
	Accuracy	*Precision*	*Recall*	*F score*	*Recall* (%)	*Recall* (%)	*Recall* (%)
FinBERT	88.2	87.2	88.5	87.8	88.5	87.3	89.7
BERT	85.0	83.6	85.1	84.2	86.3	84.1	84.8
LM Dictionary	62.1	64.4	58.0	58.1	40.2	84.3	49.5
NB	73.6	71.8	70.7	71.1	75.4	78.4	58.2
SVM	72.6	71.0	68.8	69.6	68.7	82.8	54.9
RF	71.9	75.4	64.8	66.8	70.7	86.9	37.0
CNN	75.1	74.4	71.4	72.5	73.5	83.6	57.1
LSTM	76.3	74.8	72.5	73.3	79.3	82.1	56.0

Electronic copy available at: https://ssrn.com/abstract=3910214

FinBERT-Results-Performance across sample sizes



Electronic copy available at: https://ssrn.com/abstract=3910214

FinBERT TUTORIAL

FinBERT-Tutorial

- **Step 1:** Install the transformers library from Hugging Face
- (https://huggingface.co/transformers/installation.html) and import the two required Python classes, BertTokenizer and BertForSequenceClassification, from the library. Install and import numpy.

```
from transformers import BertTokenizer, BertForSequenceClassification
import numpy as np
```

Source:*FinBERT-A Large Language Model for Extracting Information from Financial Text* by Huang, Wang and Yang 2022

FinBERT-Tutorial

- **Step 2:** Download and load the fine-tuned FinBERT sentiment classification model and the vocabulary (FinVocab) from Hugging Face.

```
finbert = BertForSequenceClassification.from_pretrained('yiyanghkust/finbert-tone',num_labels=3)
tokenizer = BertTokenizer.from_pretrained('yiyanghkust/finbert-tone')
```

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FinBERT-Tutorial

- **Step 3:** Based on 4 test sentences, estimate the likelihood of each sentiment belonging to neutral, positive and negative category respectively

```
sentences = ["there is a shortage of capital, and we need extra financing",
             "growth is strong and we have plenty of liquidity",
             "there are doubts about our finances",
             "profits are flat"]

inputs = tokenizer(sentences, return_tensors="pt", padding=True)
outputs = finbert(**inputs)[0]

# calculate each sentence's probability of belonging to a sentiment class (round to 4 decimal places)
from scipy.special import softmax
np.around(softmax(outputs.detach().numpy(), axis=1), 4)
```

```
array([[0.0034, 0.    , 0.9966],
       [0.    , 1.    , 0.    ],
       [0.    , 0.    , 1.    ],
       [0.9889, 0.0018, 0.0092]], dtype=float32)
```

FinBERT-Tutorial

- **Step 4:** Assign each sentence to the most likely category

```
labels = {0:'neutral', 1:'positive',2:'negative'}
for idx, sent in enumerate(sentences):
    print(sent, '----', labels[np.argmax(outputs.detach().numpy()[idx])])
```

```
there is a shortage of capital, and we need extra financing ---- negative
growth is strong and we have plenty of liquidity ---- positive
there are doubts about our finances ---- negative
profits are flat ---- neutral
```

LLM: APPLIED RESEARCH

LLMs & Sentiment Indicator: “Can ChatGPT Decipher Fedspeak”?

- “Can ChatGPT Decipher Fedspeak?” by Anne Lundgaard Hansen and Sophia Kazinnik, April 2023 available from following link: https://www.newyorkfed.org/medialibrary/media/research/conference/2023/FinTech/400pm_Hansen_Paper_Kazinnik_2023.pdf?sc_lang=en&hash=9B1647BD6876D0F3959C6919BA3F82DE
- Focuses on the ability of OpenAI’s GPT models to interpret Fedspeak by:
  - ***Classification of the policy stance of announcements by FOMC vs human benchmark***
  - Narrative approach to identify monetary policy shocks
  - Automating the narrative approach of Romer and Romer (1989) for the identification of macroeconomic & monetary policy shocks

LLMs & Sentiment Indicator: “Can ChatGPT Decipher Fedspeak”?

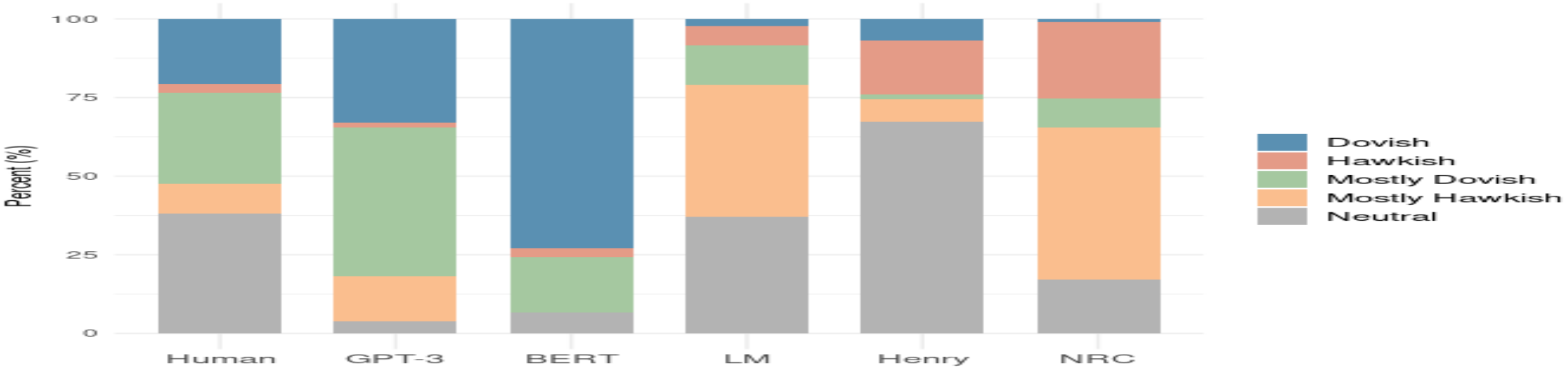
- Methodology overview:
  - FOMC minutes; 8 meetings a year; sample period 2010-2020
  - Each sentence is analyzed in isolation
  - Creating “human benchmark”: 500 sentenced picked randomly and assessed by 3 researchers
  - Classification: ”dovish”, “mostly dovish”, ”neutral”, “hawkish”, “mostly hawkish”; numerically expressed as a range from -1 to +1
  - Given sample period, more dovish than hawkish sentences

LLMs & Sentiment Indicator: “Can ChatGPT Decipher Fedspeak”?

- Methodology overview:
  - Zero-shot learning i.e. recognise new objects or concepts model has not been trained on before
  - Compares against other NLP techniques:
    - Bert
    - Lexicon based models (LM, Henry,NRC)

LLMs & Sentiment Indicator: “Can ChatGPT Decipher Fedspeak”?

Classification Problem with Zero-Shot Learning: Results



LLMs & Sentiment Indicator: “Can ChatGPT Decipher Fedspeak”?

Results with zero-shot learning

	GPT-3	BERT	LM	Henry	NRC
MAE	**0.41**	0.66	0.62	0.55	0.81
RMSE	**0.58**	0.84	0.80	0.75	0.96
Accuracy	**0.37**	0.25	0.28	0.35	0.11
Kappa	**0.18**	0.03	0.07	0.08	-0.04
	F1 score				
Dovish	**0.49**	0.31	0.07	0.17	0.04
Mostly dovish	**0.43**	0.33	0.23	0.04	0.17
Neutral	0.15	0.13	0.48	**0.57**	0.14
Mostly hawkish	**0.36**	NA	0.15	0.07	0.11
Hawkish	**0.10**	0.07	NA	0.08	0.03
	Balanced Accuracy				
Dovish	**0.71**	0.48	0.51	0.53	0.51
Mostly dovish	**0.56**	**0.56**	0.53	0.50	0.51
Neutral	0.54	0.51	**0.59**	**0.59**	0.45
Mostly hawkish	**0.67**	0.50	0.49	0.50	0.42
Hawkish	0.53	0.52	0.47	**0.56**	0.45

Results with fine-tuning

	GPT-3 (fine-tuned)	GPT-3 (zero-shot)	BERT	LM	Henry	NRC
MAE	**0.23**	0.40	0.60	0.58	0.54	0.85
RMSE	**0.40**	0.57	0.77	0.79	0.71	0.98
Accuracy	**0.61**	0.41	0.28	0.33	0.31	0.10
Kappa	**0.46**	0.21	0.01	0.15	0.00	-0.03
	F1 score					
Dovish	**0.77**	0.48	0.34	0.07	0.06	0.07
Mostly dovish	**0.53**	0.45	0.31	0.34	0.07	0.26
Neutral	**0.66**	0.24	0.18	0.58	0.52	0.04
Mostly hawkish	0.22	**0.50**	NA	0.12	NA	0.11
Hawkish	**0.80**	NA	NA	NA	NA	NA
	Balanced Accuracy					
Dovish	**0.83**	0.65	0.45	0.52	0.47	0.51
Mostly dovish	**0.67**	0.59	0.54	0.57	0.51	0.55
Neutral	**0.73**	0.57	0.53	0.67	0.52	0.40
Mostly hawkish	0.61	**0.80**	0.50	0.54	0.47	0.49
Hawkish	**0.99**	0.49	0.49	0.48	0.45	0.39

Source: Can ChatGPT Decipher Fedspeak? by Anne Lundgaard Hansen, Sophia Kazinnik :: SSRN p.28

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